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Editors
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Dear Editors:

Please consider the enclosed Research Report as a Direct Submission to *PNAS*: "Temperature, Not Structure, Sets Dynamic Arrest in an Elastin-like Protein Polymer."

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**Classification.** Physical Sciences: Applied Physical Sciences. Secondary: Biophysics and Computational Biology.

**Significance.** Whether a soft or biological material has stopped flowing is usually inferred from its structure: a scattering profile that has grown enough is read as a gel. We show that this inference can fail. In the sequence-defined elastin-like polypeptide $(VPAVG)_{30}$, four loadings held between 28.00 and 28.75 °C all assemble, but only those above 28.00 °C arrest within the hold, and at the onset of arrest the profile lies within 18% at every $Q$ of the 28.00 °C profile, which is still mobile after 6500 s. A quarter of a degree decides it, not an amount of structure. Arrest also reaches the largest length scales first, and it advances by converting material out of the mobile population rather than by slowing it: the fast-decay amplitude falls from 0.80 to 0.07 at 175 nm within an hour while the fitted diffusion coefficient varies by a factor of two.

**Novelty.** One rescaling of the waiting time, $\exp(\Delta T/T_s)$, collapses all four holds onto a single master curve with $T_s = 0.22 \pm 0.02$ K, and the same $T_s$ to within 13% in every one of the five $Q$ bins. That is a stronger statement than a rate: the sample passes through the same sequence of dynamical states at every temperature and every length scale measured, and temperature sets only the pace. A scaling temperature of a few tenths of a kelvin is far too sharp for a single contact, which places the rate-limiting step at the concerted formation of a junction in which many contacts close together.

That scaling framework is our own. One of us introduced it for thermoreversible colloidal gels [Q. Zhang *et al.*, *Phys. Rev. Lett.* **119**, 178006 (2017)], where it returns $T_s = 0.33$ and 0.45 K for two particle sizes [D. Bahadur *et al.*, *J. Chem. Phys.* **151**, 104902 (2019)]. The present work carries it from colloids to a protein polymer and finds the same cooperative regime. We cite the agreement with the nanoparticle-gel phenomenology rather than claiming it as new; what is new is that temperature and structure are separable in a protein network, and that one time scale governs every length scale we can measure.

**Why it needed this measurement.** The fast relaxation is millisecond-scale and falls at or below the shortest delay a conventional kilohertz acquisition reaches. The decomposition rests on framing X-ray speckle at 50 kHz with the XSPA-500k at APS 8-ID-I; at conventional speeds neither the length-scale ordering nor the scaling collapse is visible at all.

**Limits, stated plainly.** The 1.6 s maximum delay bounds a fraction arrested on that scale rather than an infinite-time nonergodicity factor. The scaling rests on three arresting temperatures spanning half a degree, the collapse is approximate rather than exact, and $T_s$ drifts by 13% across the five $Q$ bins.

**Companion manuscript.** A separate manuscript, in preparation from the same construct, reports the structural side: a thermally reversible transition to a hierarchical assembly carrying $\beta$-sheet-like order, and the dependence of that reversibility on chain length. The two were originally one Letter and were separated because the dynamical result stands on its own and the structural result belongs with the chain-length series. There is no overlap of figures or data between them. We are happy to supply the companion manuscript on request.

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**Data and code.** The reduced data and all analysis and figure-generation code are public at qzhang234/PubCode-2021-IDPP; every data-derived figure regenerates from that repository alone.

The manuscript is original, is not under consideration elsewhere, and has been approved by all authors. The authors declare no competing financial interest; AI-tool use is disclosed in the Acknowledgments. Suggested Editorial Board members, NAS members and reviewers are listed overleaf.

Thank you for your consideration.

Sincerely,

Qingteng Zhang, Ph.D., on behalf of all authors

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PNAS asks for a minimum of three Editorial Board members, three NAS members and three qualified reviewers. **TO BE COMPLETED BEFORE SUBMISSION.** The researchers below have relevant expertise in elastin-like polypeptides, protein and peptide assembly, small-angle X-ray scattering, and X-ray photon correlation spectroscopy, and were carried over from the earlier cover letter; they are reviewer suggestions, and the Editorial Board and NAS member lists still have to be chosen.

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